

DRAFT: The Scalar-Angular-Torsion Framework: Particle Reference Guide and Topological Taxonomy

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This document serves as the definitive particle reference guide for the Scalar-Angular-Torsion (SAT) framework. We categorize matter based on filamental persistence and topological complexity (Q) rather than standard spin-statistics. By re-architecting particles as 4D superhelical filaments within a 24-cell HSUCV lattice, we derive a unified taxonomy of True Fermions and True Bosons. We provide the scaling relations for the mass “Gear-Box” and establish the stability constraints dictated by the UV Finiteness Lock.

LAY OVERVIEW

This guide provides a comprehensive “map” of the subatomic world as seen through the SAT framework. In this view, particles are not dimensionless dots but physical, four-dimensional knots called filaments. We measure the complexity of these knots using a value called **Topological Charge (Q)**, which counts how many times filaments twist or link together. A central rule of this universe is that no more than three filaments can overlap at a single point, a limit known as the **UV Finiteness Lock**.

The framework redefines the fundamental categories of matter. A **True Fermion** is any particle that has a persistent, physical knot structure that stays intact as it moves through the universal lattice. This includes not only electrons and protons but also mesons, which were traditionally viewed as bosons. Conversely, a **True Boson** is a temporary excitation or ripple in the filaments, such as a photon or a neutrino, which lacks a permanent knot. By using the formulas in this guide, scientists can calculate the mass of any particle simply by looking at its knot geometry and the universal expansion rate.

I. INTRODUCTION TO SAT TAXONOMY

The Scalar-Angular-Torsion (SAT) framework departs from the Standard Model by defining particles as physical 4D worldline filaments following superhelical paths. Particle identity is a structural consequence of the integrated coiling history, summarized by the Topological Charge (Q) and the superhelical order (n).

II. THE PARTICLE REFERENCE TABLE

The following taxonomy categorizes particles by their topological class, link type, and persistence within the HSUCV lattice.

TABLE I. SAT Particle Taxonomy and Scaling Metrics

Class	Q	Topology	Taxonomy	Mass Gear
Baryons	3	Borromean	True Fermion	$m_B \approx \frac{Q \cdot m_0}{2B}$
Mesons	2	Hopf Link	True Fermion	Intermediate
Leptons	1	Order-1	True Fermion	$m_L = Q \cdot m_0 \cdot B^4$
Quarks	1*	Braid-Link	True Fermion	(Internal)
Neutrinos	< 1	Helix Exc.	True Boson	Pseudo-mass
Photons	0	Oscillation	True Boson	Pseudo-mass

III. TOPOLOGICAL CHARGE AND STABILITY

A. Topological Charge (Q)

Q is defined as the sum of linking and winding numbers per HSUCV node. It determines whether a filament manifests as a lepton, a baryon, or a transient boson.

- **Leptons ($Q = 1$):** Simple, single-filament Order-1 superhelices.
- **Mesons ($Q = 2$):** Persistent Hopf links involving two linked filaments.
- **Baryons ($Q = 3$):** Stable Borromean triplets of three interwoven filaments.

B. The UV Finiteness Lock

Stability is governed by the $Q \leq 3$ rule, capping the maximum number of stable filament intersections at a single node. This rule predicts the structural impossibility of $Q \geq 4$ particles, such as Lithium-4, which decays immediately due to this lattice constraint.

IV. THE MASS SCALING GEAR-BOX

Mass is reinterpreted as **Projective Resistance (R)**, the mechanical drag a 4D filament experiences as it resists the expansion of the hypersphere. The Scaling Gear-Box applies the Projection Constant ($B \approx 0.2387$ rad) to

derive the mass spectrum from the Universal Mass Anchor ($m_0 \approx 1.0073 \times 10^{-27}$ kg).

A. Baryon Scaling

For high-density triplets, mass is amplified by B acting as a divisor:

$$m_B = \frac{Q \cdot m_0}{2B} - \text{Braid-Smoothing Metric} \quad (1)$$

The Braid-Smoothing Metric accounts for the non-linear damping of resistance in high-density knots like Helium-3.

B. Lepton Scaling

For bare filaments, mass is suppressed by the fourth power of the projection distortion:

$$m_L = Q \cdot m_0 \cdot B^4 \quad (2)$$

This power-law suppression represents the minimal drag of a single non-intermeshed filament.

V. SPECIFIC PARTICLE PROFILES

A. True Fermions (Persistent Coils)

Mesons ($Q = 2$): Unlike standard models, SAT classifies mesons as True Fermions due to their persis-

tent Hopf link structure. Their internal geometric tilt is locked by the **Dirac CP Phase Lock** of exactly 270° .

Quarks ($Q = 1$ per strand): These are Order-1 coils forming 2- or 3-filament braids. They serve as the source of the strong force, reinterpreted as **Braid Rigidity**.

B. True Bosons (Transient Excitations)

Neutrinos: These are single-coil helical excitations that are “parasitic” on vacuum filaments. They lack persistent identity and exhibit flavor oscillation via precession.

Photons: Reinterpreted as rotated neutrino oscillations. Upon reaching the **Obscuration Constant** ($\theta_{obs} \approx 14.1^\circ$), they can precess out of the observable 3D slice, creating the **Dark Matter Illusion**.

VI. CONCLUSION

The SAT reference guide replaces arbitrary quantum numbers with rigid geometric invariants. By tethering all matter to the 24-cell lattice architecture, the framework ensures a zero-parameter economy across the entire particle spectrum.